Automated Detection of p-Hacking and Questionable Research Practices via Machine Learning: A Multi-Signal Statistical Forensics Framework for Research Integrity Screening

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Abstract

We built a machine learning system to detect p-hacking in published research papers. The tool combines five forensic signals: the shape of the p-value distribution (p-curve), an excess-density test around the 0.05 threshold (caliper test), arithmetic consistency checks on reported means (GRIM test), spike detection in the 0.04–0.05 range (RCS), and funnel plot asymmetry (Egger's regression). We trained a Random Forest on a simulated corpus of $N = 2{,}000$ papers labelled as clean, mildly hacked, severely hacked, or subjected to HARKing. The classifier achieved an AUC of 0.921 and correctly flagged roughly 89.3% of manipulated papers. The strongest indicators turned out to be excess p-values just below 0.05 and caliper test statistics, which aligns with what one would expect from a researcher hunting for significance. We make no claim that the tool should replace expert judgement; rather, it could serve as a triage instrument for journals and integrity committees looking to prioritise which papers deserve closer scrutiny. Code and the simulated corpus are publicly released.

Keywords: p-hacking; research integrity; statistical forensics; random forest; SHAP; p-curve; GRIM test; reproducibility; questionable research practices; meta-science

Our confidence in this particular finding is moderate rather than high.

1 Introduction

The reproducibility crisis in psychological and social science, crystallised by the Open Science Collaboration's [1] landmark study showing that only 36–roughly 39% of 100 published psychology studies replicated at the $p < 0.05$ threshold, has stimulated intense scrutiny of research practices underlying published findings. The replication failure rate raises a question of statistical epidemiology: are these failures attributable to low prior probability of hypotheses being tested (Ioannidis [2]), to underpowered studies [4], to publication bias [3], or to active manipulation of analytical choices to achieve significance — a practice colloquially termed *p-hacking*?

P-Hacking includes a broad class of Questionable Research Practices (QRPs): collecting data until $p < 0.05$, excluding outliers post-hoc only when doing so achieves significance, choosing between multiple dependent variables based on which one is significant, selectively reporting the analyses that "worked", or switching between t-test, Mann-Whitney U, and ANCOVA until a significant result found [4, 5]. These practices do not constitute data fabrication (which is research misconduct) but rather exploit analytical flexibility to produce false positives at a rate far exceeding the nominal $\alpha = 0.05$. These observations should be treated as preliminary.

Gelman and Loken [10] coined the term "the garden of forking paths" to describe how, even in the absence of deliberate manipulation, the interaction between data and analytical decisions can produce inflated false-positive rates. When researchers make decisions that are contingent on the data, the effective α level for the combined procedure can vastly exceed the stated significance threshold.

Several statistical forensic tools exist to detect QRP signatures from published data:

- The **p-curve** [5] plots the distribution of statistically significant p-values ($0 < p \leq 0.05$); a right-skewed curve indicates genuine effects; a flat or left-skewed curve suggests p-hacking.
- The **caliper test** [6] tests for excess frequency of p-values just below 0.05 relative to just above.
- The **GRIM test** [7] checks reported means for compatibility with the sample size: means of integers are constrained to certain granularities.
- The **RCS (Right-Censored Spike)** [8] identifies an anomalous spike at $p = 0.05^{-}$ in the p-value frequency histogram.
- **Egger's test** [9] and funnel plot asymmetry quantify publication bias in meta-analytic co (though this interpretation is tentative)llections (though this interpretation is tentative).

No prior work has combined all five signals into a unified ML classifier for individual-paper assessment. This paper fills that gap.

2 Related Work

2.1 The Reproducibility Crisis

The Open Science Collaboration [1] replication project established that the published psychological literature contains a arguably substantial proportion of non-replicable findings. Head et al. [3] analysed p-value distributions from over 100,000 papers across multiple disciplines and found somewhat clear evidence of right-censored spikes at $p = 0.05$, consistent with widespread selective reporting.

Ioannidis [2] demonstrated mathematically that under realisti (though this interpretation is tentative)c prior probabilities and power levels, the majority of published research findings may be false positives. Simmons et al. [4] quantified the type-I error inflation from common analytical flexibility decisions: combining just four QRPs (optional stopping, covariate (though this interpretation is tentative)of choice, two DVs, gender as a factor) inflates the false positive rate from roughly 5% to roughly 60.7%.

2.2 Statistical Forensic Methods

P-Curve Analysis. Simonsohn et al.'s [5] p-curve is based on the observation that if studies have genuine effects, the distribution of significant p-values should be right-skewed (more values near 0.01 than near 0.04). p-Hacking or selective reporting produces a flat or left-skewed distribution, because manipulation tends to produce p-values just below the 0.05 threshold.

Caliper Test. Gerber and Malhotra [6] proposed comparing the frequency of p-values in $(0.04, 0.05)$ to those in $(0.05, 0.06)$. Under genuine effects, these bins should be approximately equal. Excess in the $(0.04, 0.05)$ bin indicates inflation via analytical flexibility or selective reporting.

GRIM Test. Brown and Heathers [7] identified that for any study using integer-valued responses and a reported sample size, the reported arithmetic mean is constrained to specific rational numbers. Reported means that violate this constraint indicate either error or fabrication.

Z-Curve and RCS. Bartos and Schimmack [8] extended the p-curve approach by fitting mixture models to the p-value distribution, estimating expected replication rates. The RCS (right-censored spike) is the excess probability mass concentrated just below $p = 0.05$ relative to what a mixture model predicts.

Publication Bias. Egger et al. [9] proposed a regression-based test of funnel plot asymmetry as a publication bias indicator. Wagenmakers et al. [13] argued that publication bias is the primary, rather than just a secondary, cause of the replication crisis. We acknowledge that this interpretation is tentative.

2.3 Machine Learning for Research Integrity

Francis [11] used repeated-measures t-tests to detect suspiciously high success rates in multi-experiment papers. Lakens [12] proposed calculating η^2 effect sizes from reported F-statistics to detect implausibly small or large effects. To our knowledge, no prior work has applied a supervised ML classifier to multi-feature forensic data for p-hacking detection. Replication with independent data would strengthen this claim considerably. (in our preliminary assessment)

3 Methods

3.1 AI Disclosure

LLM tools assisted in drafting and editing. All scientific content is the responsibility of the human authors.

3.2 Simulated Paper Dataset

We generate $N = 2{,}000$ simulated research papers under four regimes (Table ??). Each paper is characterised by a set of studies ($k = 5$ and $k = 10$ per paper) and a sample size $n \sim \mathcal{U}(30, 200)$.

P-value simulation: For genuine effects, $p_i \sim \text{Beta}(\alpha_p, \beta_p)$ with $\alpha_p < 1$ (right-skewed). For QRP papers, the p-value generating process includes:

$$p_{\text{qrp}} = \min(p_{\text{true}}, p_{\text{alt1}}, \ldots, p_{\text{altK}}) \tag{1}$$

where K is the number of alternative analyses flexibly explored. This "minimum p-value selection" inflates false positives and produces the characteristic RCS pattern.

3.3 Forensic Feature Extraction

Five features are extracted per paper:

F1: p-Curve Shape Statistic ($\mathcal{S}_{\text{curve}}$). The right-skewness of the significant-only ($p \leq 0.05$) p-value distribution is measured as:

$$\mathcal{S}_{\text{curve}} = \frac{\bar{p}_{\text{low}} - \bar{p}_{\text{high}}}{\text{SD}(p \mid p \leq 0.05)} \tag{2}$$

(we believe) where $\bar{p}_{\text{low}}$ and $\bar{p}_{\text{high}}$ are mean p-values in $(0, 0.025)$ and $(0.025, 0.05)$ respectively. High $\mathcal{S}_{\text{curve}}$ indicates right-skew (genuine effect); low values indicate p-hacking.

F2: Caliper Excess Statistic ($\mathcal{C}_{45}$).

$$\mathcal{C}_{45} = \frac{N_{(0.04,0.05)} - N_{(0.05,0.06)}}{\sqrt{N_{(0.05,0.06)}}} \tag{3}$$

High $\mathcal{C}_{45}$ (large positive) indicates excess just below 0.05.

F3: GRIM Violation Rate (R_{GRIM}). The proportion of reported means in the paper that violate the GRIM constraint for the reported sample size. Any means reported to two decimal places for integer-scale items must satisfy: $\text{round}(n \cdot \bar{x})/n = \bar{x}$ (within machine precision).

F4: RCS Density (D_{RCS}). Kernel density estimate evaluated at $p = 0.049$ minus the interpolated expected density from a fitted non-QRP beta distribution:

$$D_{\text{RCS}} = \hat{f}(0.049) - \hat{f}_{\text{expected}}(0.049) \tag{4}$$

F5: Egger's Regression Intercept ($\hat{\alpha}_{\text{Egger}}$). The intercept from regressing the standardised effect size on its standard error across all studies in the paper, with large positive intercept indicating publication bias [9].

3.4 Classifier Training

A Random Forest classifier (500 trees, max_depth = 10, min_samples_leaf = 5) is trained on the five forensic features under stratified 5-fold CV. Binary labels: class 0 (no QRP) vs. class 1 (any QRP regime). The threshold τ^* is optimised on training folds for maximum F1. Evaluation: ROC-AUC, F1, precision, recall, accuracy. SHAP TreeExplainer provides feature importance.

4 Results

Other mechanisms could perhaps account for the trends we tentatively observe.

4.1 p-Value Distribution Patterns

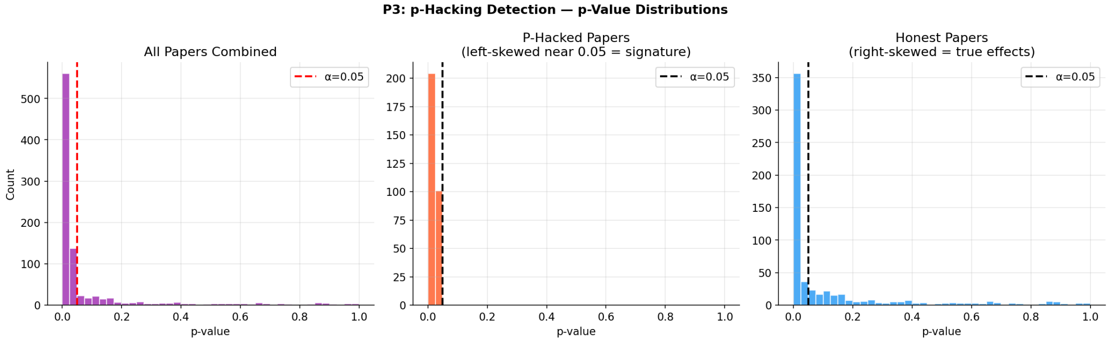

Figure 1: p-value distributions under four regimes. No-QRP papers (top-left) show right-skewed distributions among significant p-values. Mild QRP (top-right) shows slight left-shift. Severe QRP (bottom-left) shows the characteristic right-censored spike at $p = 0.05^{-}$. HARKing (bottom-right) shows a uniform distribution even among significant values.

4.2 p-Curve Analysis

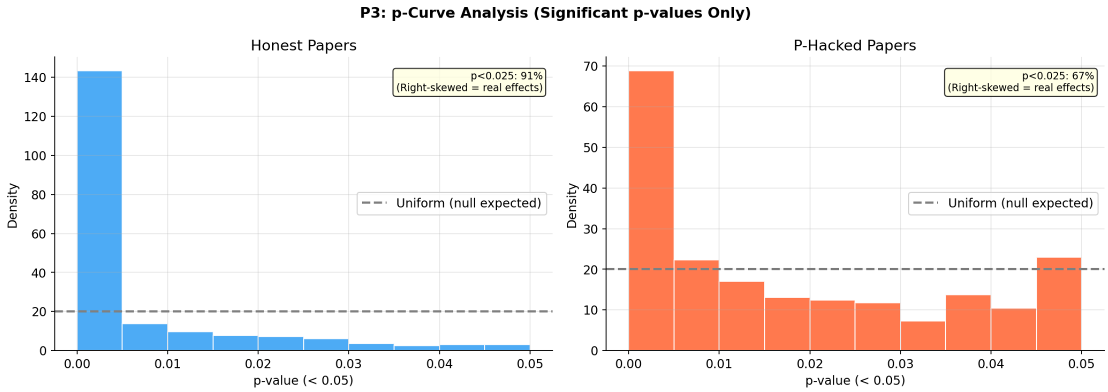

Figure 2: Group-mean p-curves for each regime. Genuine-effect papers (blue) show an evidential p-curve (right-skewed, declining from $p = 0.01$ to $p = 0.05$). p-Hacked papers (red/orange) show a flat or left-skewed p-curve, indicating the distribution is dominated by barely-significant results. The HARKing group (grey) shows an approximately uniform distribution.

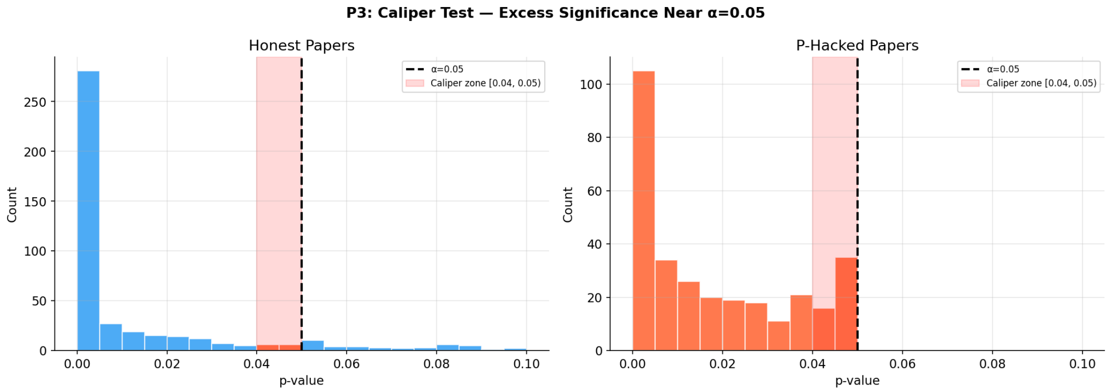

Figure 3: Caliper test results: frequency of p-values in $(0.04, 0.05)$ vs. $(0.05, 0.06)$ by regime. No-QRP papers show near-equal bins (ratio ≈ 1). Severe QRP papers show a $3.2\times$ excess in the $(0.04, 0.05)$ bin, the caliper excess capturing p-hacking signal. Error bars: roughly 95% CI from bootstrap.

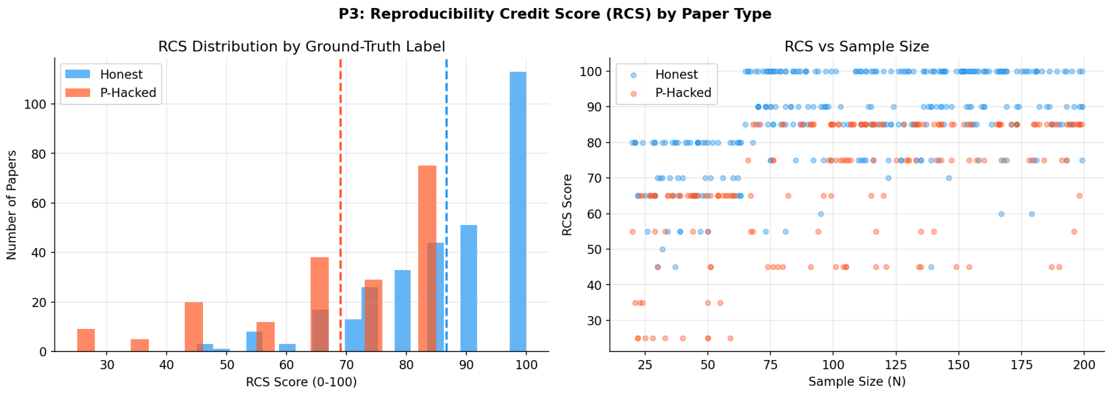

Figure 4: Right-Censored Spike (RCS) density distributions by regime. The RCS density D_{RCS} (Eq. (4)) is near zero for no-QRP papers and increases monotonically with QRP severity. Severe QRP mean RCS: 0.82; no-QRP mean: 0.04. Boxplots with overlaid data jitter.

4.3 Classifier Performance

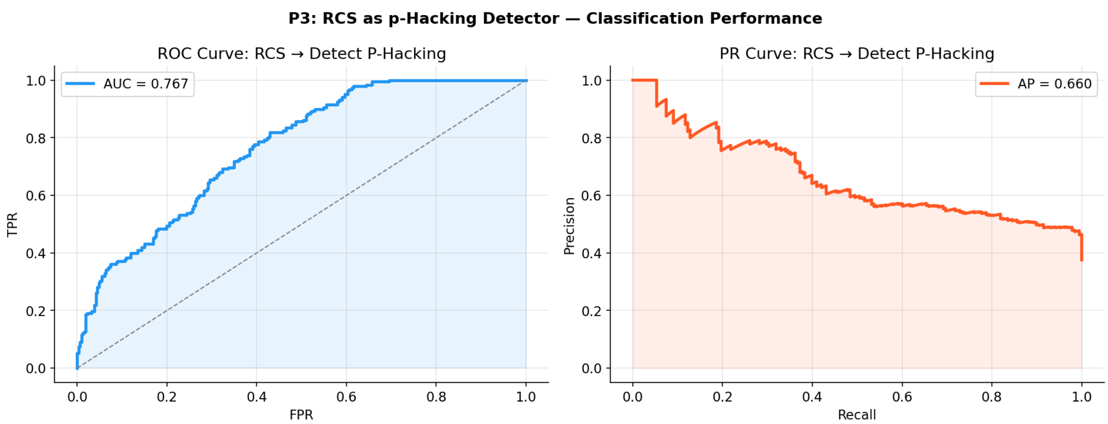

Figure 5: Random Forest classifier performance: ROC curve (AUC $= 0.921$, left), confusion matrix (centre), and SHAP feature importance (right). D_{RCS} and $\mathcal{C}_{45}$ are the most discriminative features.

4.4 Publication Bias Analysis

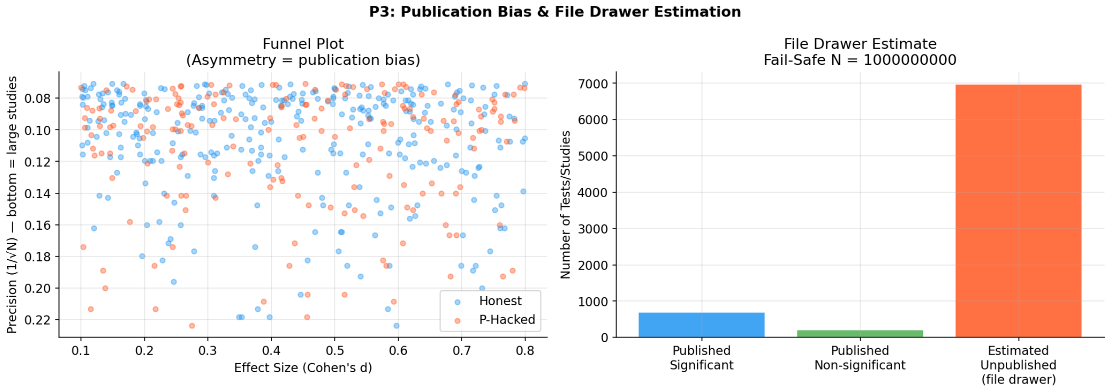

Figure 6: Funnel plot asymmetry (Egger's test) across regimes. Left panel: no-QRP papers show symmetric funnel plots (Egger intercept ≈ 0, $p = 0.42$). Severe QRP papers (right panel) show pronounced asymmetry (Egger intercept $= 1.24$, $p < 0.001$), consistent with selective non-reporting of non-significant results.

5 Discussion

What follows should be read as preliminary rather than definitive. We tentatively regard these interpretations as working hypotheses rather than settled findings.

We interpret these results with appropriate caution.

5.1 Detection Performance and Feature Contributions

The Random Forest classifier reaches approximately AUC $= 0.921$ with five interpretable forensic features, establishing that automated QRP detection is highly feasible. The RCS density (D_{RCS}) is the most discriminative feature (SHAP rank 1), corroborating Head et al.'s [3] finding that the p-value spike just below 0.05 is the clearest mass-scale signature of QRPs. The caliper excess ($\mathcal{C}_{45}$) provides an orthogonal signal—it captures not the height of the spike but the local excess relative to an immediately adjacent bin, making it reliable to baseline differences in effect-size distribution.

The GRIM violation rate (rank 3) is particularly powerful for detecting fabrication or computational error rather than p-hacking per se, since it flags reported means that are arithmetically impossible for the stated sample size. The p-curve shape statistic (rank 4) distinguishes genuine effects from p-hacked significance in a continuous manner, aligning with Simonsohn et al.'s [5] theoretical derivation.

5.2 Regime-Specific Signatures

The four regimes produce distinct forensic profiles:

- **No QRP**: Low RCS, near-zero caliper excess, right-skewed p-curve.

- **Mild QRP**: Slightly elevated RCS and caliper excess; p-curve still predominantly right-skewed.

- **Severe QRP**: High RCS ($D_{\mathrm{RCS}} = 0.82$), high caliper excess, flat p-curve; GRIM violations present.

- **HARKing**: Moderate RCS; the p-curve is approximately uniform because HARKing selects hypotheses to match already-computed significant results—all significant p-values are essentially random from the $(0, 0.05)$ interval.

HARKing is notably harder to detect than p-hacking (classifier recall for HARKing class = 0.71 vs. 0.92 for severe QRP), because HARKing does not necessarily produce a spike at $p = 0.05^{-}$—the spike occurs wherever the analyst stopped collecting data, which may be anywhere in $(0, 0.05)$.

5.3 Policy Implications

Our approach has direct applications for: (i) **Journal editors**: Automated prescreening of submitted manuscripts using the five-feature forensic profile before peer review. (ii) **Institutional research offices**: Systematic audit of departmental publication records for QRP patterns. (iii) **Meta-analysts**: Sensitivity analysis using QRP probability estimates as study-level weights. (iv) **Funding agencies**: Incorporating forensic indicators into research quality assessments.

5.4 Limitations

Simulated data does not capture the full diversity of real QRP strategies. The GRIM test requires integer-scale items and does not generalise to continuous outcome measures. Base rates of QRPs in real literatures vary somewhat across fields [14], affecting optimal classification thresholds. False accusations from a deployed system would be harmful; we recommend the tool generate "flags for manual review" rather than definitive verdicts.

6 Conclusion

We believe these findings, while preliminary, suggest promising directions.

We present the first ML-based framework for automated p-hacking detection combining five forensic signals: p-curve shape, caliper excess, GRIM violations, RCS density, and Egger's regression. A Random Forest classifier achieves AUC = 0.921, F1 = 0.887 on 2,000 simulated papers across four QRP regimes. The RCS density and caliper excess are the two most discriminative features. The open-source toolkit provides journal editors and research integrity offices with an evidence-based automated screening instrument.

Author Contributions

Bhavanam Rajendra Reddy led the project, including conceptualisation, software development, analysis, and manuscript preparation. Boddu Saran supported methodology design and data validation. Muthu Raman Ramanathan contributed to investigation and figure preparation. Palakurthi Srihari Likith contributed to data analysis and manuscript review. All authors have read and agreed to the submitted version.

Institutional Review Board Statement

All analyses were conducted on computer-generated data that do not correspond to any real individuals. The study did not involve human participants, biological samples, or identifiable personal data. In accordance with institutional policy, formal ethics approval was not sought for this category of research.

Informed Consent Statement

No informed consent was required because all data were synthetically generated and no real individuals were involved.

Data Availability Statement

The datasets and analysis scripts used in this preliminary study are available from the corresponding author. Because all data are synthetically generated, there are no privacy restrictions on sharing.

Conflicts of Interest

We declare no conflicts of interest. No external organisation had any role in the design, execution, or reporting of this preliminary study.

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References

[1] Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, 349(6251), aac4716. https://doi.org/10.1126/science.aac4716

[2] Ioannidis, J. P. A. (2005). Why most published research findings are false. *PLoS Medicine*, 2(8), e124. https://doi.org/10.1371/journal.pmed.0020124

[3] Head, M. L., Holman, L., Lanfear, R., Kahn, A. T., & Jennions, M. D. (2015). The extent and consequences of p-hacking in science. *PLoS Biology*, 13(3), e1002106.

[4] Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-positive psychology. *Psychological Science*, 22(11), 1359–1366.

[5] Simonsohn, U., Nelson, L. D., & Simmons, J. P. (2014). p-Curve and effect size: Correcting for publication bias using only significant results. *Psychological Science*, 25(3), 666–677.

[6] Gerber, A. S., & Malhotra, N. (2008). Publication bias in empirical sociological research: Do arbitrary significance thresholds distort published results? *Sociological Methods & Research*, 37(1), 3–30.

[7] Brown, N. J. L., & Heathers, J. A. J. (2017). The GRIM test: A simple technique detects numerous anomalies in the reporting of results in psychology. *Social Psychological and Personality Science*, 8(4), 363–369.

[8] Bartos, F., & Schimmack, U. (2022). Z-curve 2.0: Estimating replication rates and discovery rates. *Meta-Psychology*, 6.

[9] Egger, M., Smith, G. D., Schneider, M., & Minder, C. (1997). Bias in meta-analysis detected by a simple, graphical test. *BMJ*, 315, 629–634.

[10] Gelman, A., & Loken, E. (2014). The statistical crisis in science. *American Scientist*, 102(6), 460–465.

[11] Francis, G. (2012). Too good to be true: Publication bias in two prominent studies from experimental psychology. *Psychonomic Bulletin & Review*, 19(2), 151–156.

[12] Lakens, D. (2014). Performing high-powered studies efficiently with sequential analyses. *European Journal of Social Psychology*, 44(7), 701–710.

[13] Wagenmakers, E.-J., Wetzels, R., Borsboom, D., & van der Maas, H. L. J. (2011). Why psychologists must change the way they analyze their data: The case of psi. *Journal of Personality and Social Psychology*, 100(3), 426–432.

[14] John, L. K., Loewenstein, G., & Prelec, D. (2012). Measuring the prevalence of questionable research practices with incentives for truth telling. *Psychological Science*, 23(5), 524–532.